**What this workbook covers 本册讲什么**

Spec 4.2.1–4.2.8: the observational method with qualitative and quantitative data (tallying, event and time sampling), the six types of observation, content analysis, Freud's case study, the chi-squared test, and the use and ethics of animals in laboratory experiments, including the Animals (Scientific Procedures) Act (1986).

## How to use this booklet 怎么用

1. **Fill it in as you go** — check the answer key (last page) after each page.
2. **The six types table (p.4) and the chi-squared steps (p.7–8) matter most.** Do them from memory, under time.

## Content map 内容地图


| Page | Topic | What you must be able to do afterwards |
|---|---|---|
| **p.2** | Observation data 观察数据 | Define quant and qual data; draw a tally chart; compare time and event sampling |
| **p.3** | Types of observation 类型 | Define each of the six types with the Ainsworth example |
| **p.4** | Six types compared 对比 | Give ethics, validity and control issues for each type |
| **p.5** | Content analysis 内容分析 | Describe how to conduct one; two strengths, two limitations |
| **p.6** | Freud's case study 个案 | Describe the couch method; evaluate with subjectivity and verification |
| **p.7–8** | Chi-squared 卡方 | State the three conditions; run the nine steps; read the critical values table |
| **p.9** | Animal research 动物研究 | Give arguments for and against, including the continuity contradiction |
| **p.10** | Animal ethics law 立法 | SPA 1986, three Rs, three licences |
| **p.11** | Checkpoint + exam practice 真题 | Textbook questions: 4+2+2+3+8 marks |
| **p.12** | Answer key + brain dump 答案 | Mark honestly, then rewrite wrong answers in full |


## Revision tracker 复习打卡

☐ p.2 data ☐ p.3–4 types ☐ p.5 content ☐ p.6 Freud ☐ p.7–8 chi-squared ☐ p.9–10 animals ☐ p.11 exam

**The whole chapter in one sentence 一句话总结**

An observation **manipulates nothing**: it records behaviour as it happens, using tally charts and behavioural categories for numbers and notes for words; the six type-words trade realism against control; the chi-squared test runs on **nominal frequencies from independent groups**; and research with animals is justified only under the **1986 Act, the three Rs and three licences**.

# Observational Data 定性与定量数据

## The defining feature 与实验的分界线

In an observation there is _____ to manipulate （no independent variable）. All data is gathered by _____ （observing, listening, recording）.

## Two data types 两种数据

| Type | Definition | Textbook example |
|---|---|---|
| **Quantitative（定量）** | Can be _____ or expressed numerically （counted） | Counting how often a child picks up a _____ toy （gender-stereotyped） |
| **Qualitative（定性）** | Not numerical; concerned with _____ （words, texts, ideas, themes） | Recording what parents _____ and their non-verbal communication （say to their child） |

## The tally chart 计数表（Table 23.1）

| | Male | Female |
|---|---|---|
| **Small car** | _____ （3） | _____ （6） |
| **Large car** | _____ （4） | _____ （2） |

Categories must be _____ （clear and unambiguous）; observers need training in the _____ （operational definitions of behavioural categories）.

## Time sampling vs event sampling 两种抽样

| Technique | How it works | Its problem |
|---|---|---|
| **Time sampling（时间抽样）** | Observe at fixed _____, e.g. every 30 seconds, recording all observed （time intervals） | May not be _____: behaviours between windows are missed （representative） |
| **Event sampling（事件抽样）** | Record a _____ behaviour every time it happens （predetermined） | If too many instances happen _____, not all can be recorded （at once） |

**中文理解：两种抽样各漏什么**

时间抽样按表看钟，**漏掉间隔里发生的行为**；事件抽样盯住目标行为，**行为扎堆时来不及记全**。持续行为用时间抽样，偶发行为用事件抽样。

# Types of Observation 观察的类型

## Pair 1 · Naturalistic vs structured

**Naturalistic（自然观察）**: behaviour watched in a _____ environment where it would normally occur （natural）. Higher _____ validity because behaviour is natural and spontaneous （ecological）. But it raises issues of _____ and privacy if participants are unaware （consent）, and has limited _____ over extraneous variables （control）.

**Structured（结构观察）**: _____ are manipulated so the behaviour is displayed （variables）. Staged, usually in an environment the researcher _____, sometimes behind a one-way screen （controls）.

> **Worked example: Ainsworth's Strange Situation**
> Ainsworth et al. (1978) studied _____ （parent–child attachment）. The sequence (mother leaves, stranger enters, reunion) would _____ occur naturally （rarely）, so she staged it: a _____ observation in a controlled environment （structured）.

## Pair 2 · Participant vs non-participant

**Non-participant（非参与）**: observing from a distance; risk of _____ if the detached observer misinterprets behaviour （observer bias）. Gain: the observer can _____ on recording, collecting data more objectively （concentrate）.

**Participant（参与）**: joining the group; _____ stranger effect on participants' behaviour （no）, reveals data other methods miss. But hard to record _____ while taking part, and the researcher may lose _____ （notes; objectivity）.

## Pair 3 · Overt vs covert

**Overt（公开）**: participants know; _____ consent possible and withdrawal respected （informed）. **Covert（隐蔽）**: participants unaware, _____ problematic （ethically）; but behaviour unchanged, so more _____ （valid）.

## Combinations 组合

Textbook examples: a _____, participant, _____ observation (an anthropologist joining a tribal group) （naturalistic; overt）; or a _____, non-participant, _____ observation (secret watching behind a one-way mirror) （structured; covert）.

> **中文理解：每对选择都在换东西**
> 自然换真实（生态效度），结构换控制（因果和重复）；参与换洞察，非参与换客观；公开换伦理，隐蔽换真实。答题写成「选了 A 换来什么、付出什么」就自动成对。

## Six Types Compared 六类型对比

### Issues table 问题表（课本 Activity 2 的依据）

| Type | Ethics | Objectivity and validity | Reliability and control |
|---|---|---|---|
| **Naturalistic** | _____ issues if unaware (consent and privacy) | Real behaviour; higher _____ validity (ecological) | Limited control; hard to _____ exactly (repeat) |
| **Structured** | Consent and withdrawal possible | Behaviour may be _____ by awareness; low eco validity (modified) | Control and _____; replicable (manipulation) |
| **Participant** | Usually overt, so consent possible | Reveals hidden data, but researcher may lose _____ (objectivity) | Hard to record _____ (notes) |
| **Non-participant** | May still be noticed | _____ bias: detached observer misinterprets (Observer) | Can concentrate on recording; more _____ (objective) |
| **Overt** | _____ consent; right to withdraw (Informed) | Participants may change behaviour when watched | Systematic open recording |
| **Covert** | Participants unaware: _____ problematic (ethically) | Behaviour unchanged, so more _____ (valid) | Recording must be discreet |

### Exam tip 考试提醒（课本原文）

**Outline + example**

Be ready to outline each type **and give examples**. _____ state which type you are describing (Explicitly). Issues that apply to one type may not apply to another: participant observations affect the behaviour observed, but that may not be true of _____ _____ observations (covert non-participant).

# Content Analysis 内容分析

## What it is 定义

Content analysis observes the _____ of certain words, images or concepts within material （incidence）, typically in _____, books, films, newspapers （advertisements）. Example from the textbook: instances of _____ in children's books （gender stereotyping）. It is therefore a form of _____ observation （indirect）.

## How to conduct one 怎么做

1. Categories _____ in advance （agreed） (stereotypical and non-stereotypical behaviour by males and females).
2. _____ the number of times each category occurs （Tally）.
3. Qualitative analysis can examine _____ of words, concepts and pictures, then make inferences （meanings and relationships）.
4. Content must be _____ into clear and manageable categories （coded）.

## Evaluation 评价

| | Point |
|---|---|
| **Strength 1** | _____ method: data from existing sources, rarely ethical issues; still respect _____ （Unobtrusive; confidentiality） |
| **Strength 2** | _____ easily assessed: replicable using the same categories; useful for _____ material and trends over time （Reliability; historical） |
| **Weakness 1** | Relies on the personal _____ of the researcher (subjectivity); _____ validity risk if categories do not measure what they intend （interpretation; internal） |
| **Weakness 2** | Purely _____: shows what is there but not _____ （descriptive; why）. Limited by _____ of material （availability）. Media trends may not reflect reality (dramatic events get more coverage); findings may be _____ if one culture （culture-biased） |

**中文理解：内容分析的三句话**

一句定义：数别人 produced 的材料里的类目，间接观察。一句优点：材料现成不打扰人，同类目可复制所以信度好查。一句缺点：怎么数靠研究者主观，数出来只有 what 没有 why。

# Freud's Case Study 弗洛伊德的个案研究

## Definition 定义

A case study is a detailed investigation into _____ （one individual or a small group）, used when someone is _____ or has experienced something that could not be studied in a laboratory （unique/rare）. Many methods can be combined: _____, interviews, experiments （observations）.

## What Freud did 弗洛伊德的做法

- _____ and dream analysis to uncover unconscious wishes （Free association）.
- Clients lay on a couch _____ him, to encourage talking without censorship （facing away from）.
- _____: when a client says something they did not mean, the unconscious revealing itself （Freudian slips）.
- Analysed dreams and looked for _____ and counter-transference （transference）.
- Analysed the _____ data and gave his interpretation to the client （qualitative）. His case studies informed his theories and were also used as _____ （a therapy）.

## Evaluation 评价

| | Point |
|---|---|
| **Strength 1** | Sometimes the _____ way to gather information: Freud could not have caused hysteria in a client （only） |
| **Strength 2** | High _____ validity: data from the individual in a natural setting; rich detail used to treat patients and develop theories （ecological） |
| **Weakness 1** | Only _____ individuals, typically _____ Viennese women, so hard to generalise （unique; middle-class） |
| **Weakness 2** | Different analysts find _____ meanings (subjective); the _____ cannot be directly observed, so not empirically _____ （different; unconscious; verified） |

**中文理解：优点缺点出自同一套做法**

他躺在沙发上的做法换来了细致的信息（自然情境、细节丰富、边研究边治疗）；主观也是这套流程带来的（不同分析师解出不同含义，潜意识又无法实证）。评价时把这条因果链写出来就是高级 AO3。

# Chi-Squared Test 卡方检验（一）

## Three conditions 三个入场条件（必背）

1. The hypothesis predicts a _____ or an _____ （difference; association）.
2. The level of measurement is _____ （nominal）.
3. The design uses _____ measures: at least two different groups （independent）.

**Exam tip（课本原文）**

A clue that chi-squared is needed: _____ are totalled （frequencies）. It is not used when participants' _____ scores are known （individual）. Do not convert numbers into _____ or percentages （averages）.

## Worked example: toy choice 玩具选择例题

| | Stereotypical toy | Non-stereotypical toy | Row total |
|---|---|---|---|
| **Girls** | _____ (A) （8） | _____ (B) （12） | _____ （20） |
| **Boys** | _____ (C) （17） | _____ (D) （3） | _____ （20） |
| **Column total** | _____ （25） | _____ （15） | _____ （40 overall） |

## Steps 1–4 前四步

| Step | Rule | This example |
|---|---|---|
| **1. Contingency table** | Put _____ frequencies in （observed） | 8, 12, 17, 3 |
| **2. Expected frequency** | E = _____ total × _____ total ÷ _____ total （row; column; overall） | A: 12.5 · B: 7.5 · C: 12.5 · D: 7.5 |
| **3. Subtract** | _____ – _____ for each cell （O – E） | –4.5 · +4.5 · +4.5 · –4.5 |
| **4. Square** | (_____)² for each cell （O – E） | _____ in every cell （20.25） |

## Steps 5–9 后五步

| Step | Rule | This example |
|---|---|---|
| **5. Divide by E** | $(O - E)^2 \div$ _____ for each cell （E） | A: 1.62 · B: 2.7 · C: 1.62 · D: 2.7 |
| **6. Add up** | Sum = _____ value of $\chi^2$ （observed） | 1.62 + 2.7 + 1.62 + 2.7 = _____ （8.64） |
| **7. Degrees of freedom** | df = (_____ − 1) × (_____ − 1) （rows; columns） | (2−1) × (2−1) = _____ （1） |
| **8. Critical value** | Look up df and _____ level, noting one- or two-tailed （significance） | One-tailed, p = 0.05, df = 1 → _____ （2.71） |
| **9. Compare** | Observed _____ critical → significant, reject the _____ hypothesis （≥; null） | _____ （8.64 > 2.71: significant） |

## Critical values table 临界值表（Table 23.3）

| df | One-tailed test | | | Two-tailed test | | |
|---|---|---|---|---|---|---|
| | **0.05** | **0.025** | **0.01** | **0.10** | **0.05** | **0.02** |
| **1** | _____ （2.71） | 3.84 | 5.41 | 2.71 | _____ （3.84） | 5.41 |
| **2** | 4.60 | 5.99 | 7.82 | 4.60 | 5.99 | 7.82 |

**中文理解：九步口诀**

列表 → 算期望 → 相减 → 平方 → 除以期望 → 加总 → 算自由度 → 查表 → 比大小。考试最爱三小问：为什么选卡方（三条件）、自由度（行减一乘列减一）、显著性判断（观测值大于临界值就显著，拒绝虚无假设）。

## Practice 自测

*A study with df = 2 produces an observed $\chi^2$ of 6.31. Is this significant at $p \le 0.05$, two-tailed?*

# Animal Research: For and Against 动物研究正反方

## Background 背景

Learning theorists such as _____ and _____ built their methods on animal experimentation （Pavlov; Skinner）, manipulating an IV to see its effect on behaviour.

## Arguments in favour 支持方（四条）

1. **Breeding speed:** a rat's gestation period is typically _____ days （22）; large samples bred quickly; generational effects appear without long waits.
2. **Control:** animals can be _____ in ways humans cannot （caged）; variables isolated, e.g. the _____ box dispensing food on a lever press （Skinner）.
3. **Fewer ethical limits:** unnecessary pain must be avoided but _____ and distress are permitted （pain）; research still heavily _____, with regular welfare _____ （regulated; inspections）.
4. **Benefits and continuity:** insight into drug treatments, transplants, surgery, _____ （cloning）; humans share _____ ancestry with other animals, especially primates （common/evolutionary）, so we can generalise to some extent. The _____ argument: a small number of animals suffering is justified because it helps a significant number of people （utilitarian）; _____ (1991) supports this when research furthers the cause （Gray）.

## Arguments against 反对方（四条）

1. **Credibility:** lab animal behaviour is _____ from natural behaviour, so eco validity is low （different）.
2. **Too many differences:** evolution, genes, cognition, emotion; human behaviour has more variables (culture, social norms, _____) （language）; humans have a proportionally much larger _____ （cortex）. A newborn mouse equals a _____ （seven-week-old foetus）.
3. _____: mistakenly attributing human qualities to animals （Anthropomorphism）; and the continuity contradiction: if animals are so similar, why not the same _____ considerations? （ethical）.
4. _____: Singer (1975) called giving human suffering priority a form of discrimination （Speciesism）; benefits are unknown until _____ the study （after）; on this view _____ regulation can justify animal research （no amount of）.

**中文理解：连续性是双刃剑**

支持方靠连续性（共同祖先、相似神经系统）说结果能推广；反方用同一把刀反击：既然这么相似，凭什么不给同样的伦理保护？再一步就是 Singer 的物种歧视。把这条正反两用写出来比堆理由高级。

### Legislation 立法

- The **Animals (Scientific Procedures) Act (_____)** covers all animal research in the UK （1986）. It covers any scientific procedure that may cause _____, suffering, distress or lasting harm to a 'protected' animal （pain）.
- Protected animals = all _____ _____ plus a single invertebrate species （non-human vertebrates; the octopus）.
- Also follow the **Animal Welfare Act (_____)** （2006）; in Europe, _____ governs national regulations （Directive 2010/63/EU）.

### BPS position and the three Rs 三个 R

BPS: permission will not be granted unless the researcher can _____ the costs to the animals in relation to the likely _____ （justify; benefits）. Curiosity alone is not enough. Three Rs (Russell and Burch, _____) （1959）:

| R | Meaning |
|---|---|
| **Replace（替代）** | Use _____ alternatives whenever possible （non-sentient） |
| **Reduce（减少）** | Use _____ animals （fewer） |
| **Refine（优化）** | Refine procedures to _____ suffering （minimise） |

### The three licences 三张许可（three Ps）

| Licence | What it regulates |
|---|---|
| **Project licence** | Each project gets a new licence specifying the _____ and _____ of animals （species; number） |
| **Personal licence** | For each researcher, only after _____ （training） |
| **Place licence** | Regulates the _____ in which animals are kept; regularly inspected （conditions/housing） |

**中文理解：三 R 管怎么对待，三证管谁批准**

Replace、Reduce、Refine 管**怎么对待动物**（换掉、减量、减痛）；Project、Personal、Place 管**谁批准谁来做在哪做**。锚点细节：**1986 法案 + 八爪鱼是唯一受保护的无脊椎动物**。

## Checkpoint（课本）

**1.** Match: 1 Maya waited outside a cinema recording who went in · 2 Leo secretly watched students in a common room · 3 Asha announced she would watch who students sit next to · 4 Benito staged a play area in a lab · 5 Carlos joined a group of bikers · 6 Jamal watched from a distance.

1 _____ 2 _____ 3 _____ 4 _____ 5 _____ 6 _____

**2.** Name the three Rs and the three licences (three Ps).

**3.** Choose the three conditions for using a chi-squared test: independent groups · ordinal data · test of difference or association · interval data · repeated measures · nominal data.

## Exam practice 真题（课本）

*1. Niro observed how many visitors chose a glass walkway or an avoiding path at the top of a tall building. Explain one strength and one weakness of Niro using an observation research method in this study. (4)*

*2. Angelo tallied toys played with by nursery children (boys 18/32; girls 34/12). (a) Justify the use of a chi-squared test. (2) (b) Observed value 13.87, df = 1, two-tailed, $p \leq 0.05$. Is it significant? Explain. (2)*

*3. Evuska observed mobile phone use by drivers aged 25 or below vs 50 and above, using time sampling. Describe how she carried out the observation using time sampling. (3)*

*4. Evaluate content analysis as a research method. (8)*

# Answer Key 答案速查 + Brain Dump

## Answer key 答案速查

**p.2** no IV · counted / gender-stereotyped · words etc. · 3/6/4/2 · clear and unambiguous · operational definitions · time intervals; representative; predetermined; at once.
**p.3** natural · ecological · consent · control · variables · controls · parent–child attachment · rarely · structured · observer bias · concentrate · no · notes · objectivity · informed · ethically · valid · naturalistic/overt · structured/covert.
**p.4** consent and privacy · ecological · repeat · modified · manipulation · objectivity · notes · Observer · objective · Informed · ethically · valid · Explicitly · covert non-participant.
**p.5** incidence · advertisements · gender stereotyping · indirect · agreed · Tally · meanings and relationships · coded · Unobtrusive/confidentiality · Reliability/historical · interpretation/internal · descriptive/why · availability · culture-biased.
**p.6** one individual or small group · unique/rare · observations · Free association · facing away from · Freudian slips · transference · qualitative · a therapy · only · ecological · unique/middle-class · different/unconscious/verified.
**p.7–8** difference; association · nominal · independent · frequencies · individual · averages · 8/12/17/3 · 20/20/25/15/40 · observed · row, column, overall · O–E · O–E · 20.25 · E · 8.64 · rows, columns · 1 · significance · 2.71 · ≥, null · 8.64>2.71 significant · 2.71 / 3.84 · practice: 6.31 > 5.99, significant at $p \leq 0.05$ two-tailed with $df = 2$.
**p.9** Pavlov, Skinner · 22 · caged · Skinner · pain · regulated, inspections · cloning · common/evolutionary · utilitarian · Gray · different · language · cortex · seven-week-old foetus · Anthropomorphism · ethical · Speciesism · after · no amount of.
**p.10** 1986 · pain · non-human vertebrates, the octopus · 2006 · Directive 2010/63/EU · justify, benefits · 1959 · non-sentient · fewer · minimise · species, number · training · conditions/housing.
**p.11** 1 naturalistic 2 covert 3 overt 4 structured 5 participant 6 non-participant · three Rs: replace, reduce, refine; three Ps: project, personal, place · nominal data, independent groups, test of difference or association.

## Model answer highlights 高分要点（4+8 分题）

### Q1 玻璃走道（4 分）

Strength (ecological validity): real setting where the behaviour normally occurs, so path choice reflects genuine avoidance of heights. Weakness (lack of control): weather, crowd levels and individual differences could not be controlled, so fear of heights may not be the only cause and the observation cannot be repeated exactly.

### Q4 内容分析（8 分骨架）

Two strengths: unobtrusive (existing sources, few ethical issues, historical material available) + reliability (same categories make replication easy). Two weaknesses: subjective (personal interpretation; internal validity if categories misfit) + descriptive only (what not why; limited by material availability; media ≠ reality; culture bias). Conclusion: valuable where live research is impossible; check across coders and cultures.

## Brain dump 空白重写

Rewrite, from memory and in full, any answer you got wrong or left blank above: